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📊 h-index 12

📊 i10-index 14

About

Dr. Abderraouf Maoudj is a roboticist specializing in next-generation robotic manipulation and learning systems. With more than a decade of combined academic and industrial R&D experience across Denmark, Algeria, and Saudi Arabia, he translates rigorous theory into field-tested robotic platforms in challenging real-world environments.

Currently a Post-Doctoral Research Fellow at the Interdisciplinary Research Center for Intelligent Manufacturing & Robotics (IRC-IMR), KFUPM, he leads research on lifelong learning, multi-arm coordination, and human-robot collaboration in dynamic, safety-critical settings. His vision is to enable robots that can:

- Learn new manipulation skills from just a handful of human or video demonstrations.
- Adapt entirely offline using learned world models to “dream” possible actions without physical interaction, an emerging approach we are developing using a Recurrent State-Space Model (RSSM) to fine-tune diffusion-based robotic skills with reinforcement learning. This framework enables safe, sample-efficient policy improvement and addresses the limitations of imitation learning that often cause failures in unseen scenarios.
- Share and execute skills seamlessly across multi-robot networks, guided by natural language and vision-language-action (VLA) models.

Earlier roles include a three-year postdoctoral position at the University of Southern Denmark and seven years (Senior Researcher) at Algeria’s Centre for the Development of Advanced Technologies (CDTA), where Dr. Abderraouf consistently translated academic breakthroughs into deployable, factory-floor solutions. At both SDU and CDTA, his research focused on enabling large teams of autonomous robots to coordinate and complement each other to perform complex tasks through real-time, localized decision-making. By leveraging distributed control and continuous learning, he has developed advanced methods to optimize multi-robot fleet coordination in warehouses and fulfillment centers, critical infrastructure that forms the backbone of modern e-commerce and logistics. These innovations improve operational efficiency, reduce costs, and enhance competitiveness, outcomes that are vital for nations striving to upgrade their manufacturing capabilities and strengthen their position in global supply chains. This work significantly impacts automated warehousing and smart manufacturing, enabling robust, adaptable systems for next-generation industrial automation.

Dr. Maoudj has spearheaded projects with industrial partners across Denmark, Algeria, and Saudi Arabia—including Siemens, Enable Robotics, Mobile Industrial Robots (MiR), Sonacome, and Aramco (developed robotic pipe-cleaning systems)—delivering AI-enabled solutions for manufacturing, logistics, and inspection.

His approach combines advanced control theory, machine learning, and systems engineering to create scalable, resilient, and safe robotic systems. This blend of vision, technical depth, and cross-sector collaboration positions his work at the forefront of autonomous skill learning and human-multi-robot collaborative intelligence for applications from advanced manufacturing to safety-critical and hazardous domains.

Education

2013 – 2018: Ph.D. Process Control & Robotics, University of Sciences and Technology Houari Boumediene (USTHB).

Thesis: Remote Control of a Heterogeneous Multi-Robot System: application to the RobuTER/ULM and AGV-M6 platforms.

2010 – 2012: M.Sc. Electronics & Control, Mohamed Seddik Ben Yahia University, Jijel.
Thesis: Design and Realisation of a Mobile Robot.

2007 – 2010: B.Sc. Electrical Engineering, Mohamed Seddik Ben Yahia University, Jijel.
Capstone: Development of a Card Reader.

Professional Experience

2023–Present Post-Doctoral Fellow, Interdisciplinary Research Center for Intelligent Manufacturing & Robotics (IRC-IMR), KFUPM.
Research focus: lifelong learning and adaptive control for collaborative robots—developing transformer-based multi-arm coordination, vision–language–action models, and zero-shot task generalisation that enable robots to perform previously unseen tasks without explicit training. Established a robotics test-bed for research and teaching. candidates.

2020–2023 Post-Doctoral Researcher, SDU Biorobotics, Denmark.
Research focus: Decentralized/Distributed multi-agent planning for e-commerce logistics; developed frameworks that deliver sub-second coordination for more than one thousand robots in dynamic warehouses, significantly enhancing throughput efficiency.

2014–2020 Senior Researcher, Robotics Division, CDTA, Algeria.
Research focus: Led innovations in Industry 4.0 technologies, cyber-physical systems integration, distributed multi-agent control architectures, and smart automation solutions for manufacturing environments.

Selected R&D Projects

2023–Present Advancing Robotics for Industry 4.0 — Post-doctoral Researcher, IRC-IMR, KFUPM.
Leading a strategic research programme that redefines how collaborative robots learn, adapt, and execute complex tasks in dynamic, safety-critical environments. My vision is to enable robot teams that can learn a new skill from just a handful of human or video demonstrations, adapt entirely offline by “dreaming” in a learned world model, and deploy those capabilities across a multi-robot network without additional physical training. This approach ensures safe, efficient skill acquisition for hazardous or high-cost domains such as nuclear fusion maintenance, offshore energy, and advanced smart manufacturing.

- **Current focus — multi-arm manipulation & coordinated motion control:**
Developing transformer-based inter-arm attention mechanisms that predict partner intent, allocate subtasks, and retime trajectories, all while enforcing formal safety envelopes for human–robot collaboration in confined and high-risk workspaces. New videos of my work are available on our lab’s channel: [video](#)
- **Near-term roadmap — few-shot, transferable skill learning:**
Building a lifelong and transfer learning pipeline that distills manipulation policies from kinesthetic demonstrations, web videos, and simulation roll-outs; integrates reward learning directly from human video; and bridges the sim-to-real gap using diffusion-policy world models for fully offline adaptation.
- **Long-term vision — human–multi-robot collaborative intelligence:**
Establish a *Vision–Language–Action (VLA)* foundation model that fuses CLIP-style visual embeddings, large language model semantics, and proprioceptive states for zero-shot task generalisation, natural language planning, and safe skill sharing across robot networks in industrial and remote operations.

This programme delivers an open skill repository, secure communication protocols, and an adaptive-control stack deployable to any compliant manipulator or mobile platform. It advances national strategic goals such as *Saudi Vision 2030* and has direct industrial relevance, with collaborations spanning Denmark, Algeria, and Saudi Arabia, including work with Aramco on robotic pipe-cleaning systems.

2021 – 2023: Swarm Robotics for Industry 4.0 — Post-Doctoral Researcher, SDU.

Centering on the distributed control of intelligent robot fleets pertinent to Industry 4.0, this research developed efficient coordination methods for large fleets in industrial settings. In this project, I tackled a pressing bottleneck in modern fulfillment centres: how to move thousands of shelves, parcels, and totes without building an equally large data centre in the back office. By developing distributed control algorithms and continuous learning frameworks, advanced methods have been developed to optimize multi-robot fleet coordination in warehouses and fulfillment centers, which are critical to modern e-commerce and logistics infrastructure. These innovations enhance operational efficiency, reduce costs, and improve scalability, delivering competitive advantages essential for nations seeking to advance their manufacturing capabilities and strengthen global supply chain resilience..

- *Capacitated multi-agent control/scheduling framework*: Aiming to boost efficiency in logistics, warehouses, and transportation sectors, I designed and implemented a fully decentralized capacitated multi-agent control and scheduling framework specifically targeting logistics optimization. This system enables each autonomous guided vehicle (AGV) to reason locally while still converging on fleet-level optimal solutions, accommodating product compatibility constraints and vehicle capacity limitations.

- *Swarm robots motion coordination in dynamic industrial environments and fulfillment centers*: My most significant contribution was developing a fast, scalable, and robust decentralized motion planner capable of coordinating up to a thousand mobile robots in large warehouse environments. This innovation has a direct economic impact on modern e-commerce operations—the backbone of today’s retail industry—by dramatically increasing order fulfillment capacity while minimizing capital expenditure. The system successfully bridges the gap between theoretical simulation and real-world deployment by enhancing robot negotiation protocols and conflict resolution mechanisms in dynamic, unpredictable environments.

- Released the open-source ANTS planner and Benchmarks for research and teaching ([GitHub](#), [video](#)); adopted by academia and industry.

2014 – 2016: Remote Control of Heterogeneous Robots — Robotic Engineer|PhD student, CDTA.

As part of my doctoral research, this project involved:

- Developing a distributed multi-agent system for controlling industrial robot fleets.
- Implemented DRL-driven path optimization and fuzzy-logic controllers for mobile manipulators.
- Developing a fault-tolerant leader–follower control system for swarm mobile robots that maintains formation integrity under single-robot failures

2016 – 2020: Industry 4.0 Platform Development — Senior Researcher, CDTA.

This project entailed designing and implementing advanced algorithms and control systems for Industry 4.0 platforms, seamlessly integrating robotic innovations with smart automation principles. Key responsibilities included:

- Integrated KUKA LBR iiwa with Java/ROS middleware for collaborative assembly.
- Implemented an RFID-based part-tracking system linked to Siemens PLCs, boosting inventory accuracy.
- Upgraded the facility’s AGV fleet with a deep reinforcement learning navigation system that align with the control architecture of the I4.0 platform.

2016 – 2018: Aquatic Boat Robot for Bathymetry — Senior Researcher, CDTA.

This project aimed to provide a secure solution for bathymetric measurement using a remotely controlled boat robot for the Algerian National Hydrological Agency. My key role in this project focused on the electronic realization and the control of the robot.

Teaching and Supervision

Teaching and mentorship have been integral to my academic journey. Over the years, I have conceived and delivered undergraduate and graduate courses—including *Introduction to Robotics and AI*, *Electronics for Robotics*, *PLC Automation*, *Autonomous Navigation*, and *Applied Robotics*—that blend simulator work and physical-robot labs. My dual experience in academia and industry enables me to lead research initiatives and ensure that students are well-prepared for the complexities of modern robotics applications, so that they gain problem-solving skills relevant to emerging industry needs.

Skills

Robotics: Python/C++, ROS 1/2, KUKA LBR iiwa, MiR, Enable Robotics MM, Meca500.

Control Electronics: Sensor conditioning; STM32/Arduino; Siemens TIA-Portal; real-time PLC control.

Software: Python, C/C++, Java, C#, MATLAB, R; Git, Docker, Linux.

Languages: Arabic (native), English (fluent), French (fluent).

Selected Publications

Journal Articles

1. A. Maoudj, S. Mekid, M. Nouioua, “ML-BiRRT: Machine-learning-enhanced bi-directional RRT for multi-arm robot motion planning and coordination,” *Robotics and Autonomous Systems*, under review, 2025.
2. A. Maoudj, A. L. Christensen, “Improved decentralized cooperative multi-agent path finding for robots with limited communication,” *Swarm Intelligence*, 2023.
3. A. Maoudj, A. Kouider, A. L. Christensen, “The capacitated multi-AGV scheduling problem with conflicting products: model and a decentralized multi-agent approach,” *Robotics and Computer-Integrated Manufacturing*, 81:102514, 2023.
4. A. Maoudj, A. Hentout, “Optimal path planning based on a Q-learning algorithm for mobile robots,” *Applied Soft Computing*, 97:106796, 2020.
5. U. Farhadi, H. Hess, A. Maoudj, A. L. Christensen, “Evolution of path costs for efficient decentralized multi-agent path finding,” *Swarm and Evolutionary Computation*, 93:101833, 2025.
6. B. Mouad, A. Maoudj, A. L. Christensen, “HM-DRL: Enhancing multi-agent path finding with a heat-map-based heuristic for distributed deep reinforcement learning,” *Applied Intelligence*, in press, 2025.
7. B. Mouad, A. L. Christensen, A. Maoudj, “MAPF-Relocate: Mitigating Congestion through Strategic Agent Relocation in Multi-Agent Pathfinding,” *Autonomous Robots*, under review, 2025.
8. A. Kharidege, A. Maoudj, “Integration of a vision system with a robotic polishing system for machining a variety of workpieces,” 2024.
9. A. Hentout, A. Maoudj, M. Aouache, “A review of fuzzy-logic approaches for collision-free path planning of manipulator robots,” *Artificial Intelligence Review*, 56(4):3369–3444, 2023.
10. A. Hentout, A. Maoudj, A. Kouider, “Efficient fuzzy-logic control of indoor mobile robots evolving in static and dynamic workspaces,” 2023.
11. A. Hentout, A. Maoudj, A. Kouider, M. Aouache, “Effective GA approach for municipal solid-waste collection: dynamic capacitated vehicle routing in smart cities,” *International Journal of Artificial Intelligence and Soft Computing*, 7(4):329–352, 2022.
12. A. Hentout, M. Aouache, A. Maoudj, I. Akli, “Human–robot interaction in industrial collaborative robotics: a literature review of the decade 2008–2017,” *Advanced Robotics*, 33(15–16):764–799, 2019.
13. A. Hentout, A. Maoudj, D. Guir *et al.*, “Collision-free path planning for indoor mobile robots based on rapidly-exploring random trees and PCHIP,” *International Journal of Imaging and Robotics*,

19(3), 2019.

14. A. Hentout, A. Maoudj, M. Ouahal, A. Gasmi, “Fault-tolerant multi-agent scheme for leader/follower formation control of homogeneous mobile-robot teams,” *International Journal of Systems, Control and Communications*, 10(2):95–125, 2019.
15. A. Hentout, A. Maoudj, D. Yahiaoui, M. Aouache, “RRT-A*-BT approach for optimal collision-free path planning for mobile robots,” *Algerian Journal of Signals and Systems*, 4(2):39–50, 2019.
16. A. Maoudj, B. Bouzouia, A. Hentout, A. Kouider, R. Toumi, “Distributed multi-agent scheduling and control system for robotic flexible assembly cells,” *Journal of Intelligent Manufacturing*, 30:1629–1644, 2019.
17. A. Maoudj, A. Hentout, B. Bouzouia, R. Toumi, “On-line fault-tolerant fuzzy-based path planning and obstacle avoidance for manipulator robots,” *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, 26(5):809–838, 2018.
18. A. Bakdi, A. Hentout, H. Boutami, A. Maoudj, O. Hachour, B. Bouzouia, “Optimal path planning and execution for mobile robots using a genetic algorithm and adaptive fuzzy-logic control,” *Robotics and Autonomous Systems*, 89:95–109, 2017.
19. A. Hentout, A. Maoudj, N. Kaid-Youcef, D. Hebib, B. Bouzouia, “Distributed multi-agent bidding-based approach for collaborative mapping of unknown indoor environments,” *Journal of Intelligent Systems*, 29(1):84–99, 2017.

Refereed Conference Papers

1. K. Amara, W. Dib, A. O. Djekoune, A. Maoudj, N. Zenati, “AquaRob: an unmanned surface drone for coastal surveillance and environmental monitoring,” in *Proc. 4th Int. Conf. on Embedded & Distributed Systems (EDiS)*, IEEE, 2024, pp. 193–198.
2. A. Maoudj, S. N. Mekid, A. L. Christensen, “Lifelong MAPF for shipment preparation in stochastic warehouses,” in *Proc. IEEE CASE*, 2024, pp. 1006–1011.
3. A. Maoudj, A. L. Christensen, “Decentralized multi-agent path finding in dynamic warehouse environments,” in *Proc. 21st ICAR*, IEEE, 2023.
4. A. Maoudj, A. L. Christensen, “Decentralized MAPF in warehouse environments for fleets of robots with limited communication,” in *Proc. Int. Conf. on Swarm Intelligence*, Springer, 2022, pp. 104–116.
5. M. Gaham, M. Kadri, A. Maoudj *et al.*, “Multi-agent control of an Industry 4.0 cyber-physical assembly cell,” in *Nat. Conf. on Applied Computing and Smart Technologies*, 2021.
6. A. Maoudj, A. L. Christensen, “Q-learning-based navigation for mobile robots in continuous and dynamic environments,” in *Proc. IEEE CASE*, 2021, pp. 1338–1345.
7. A. Maoudj, A. Hentout, A. L. Christensen, A. Kouider, “A fast constructive path-planning algorithm for mobile-robot navigation,” in *Proc. IEEE ETFA*, 2021, pp. 1–8.
8. A. Hentout, A. Mustapha, A. Maoudj, I. Akli, “Key challenges and open issues of industrial collaborative robotics,” in *Workshop on Human-Robot Interaction: from Service to Industry (HRI-SI)*, 2018.
9. A. Hentout, A. Tiberkak, A. Maoudj *et al.*, “Virtual pheromone-based object-searching in RFID-enabled cyber-physical robotic systems,” in *Proc. ICASS*, 2018, pp. 1–7.
10. A. Hentout, A. Beghni, A. B. Nourine, A. Maoudj, B. Bouzouia, “Mobile-robot RBPF-SLAM with LMS sensor in indoor environments,” in *Proc. ICEE-B*, 2017, pp. 1–6.
11. A. Hentout, A. Maoudj, B. Bouzouia, “A survey of development frameworks for robotics,” in *Proc. ICMIC*, 2016, pp. 67–72.
12. A. Maoudj, B. Bouzouia, A. Hentout, A. Kouider, R. Toumi, “Distributed multi-agent scheduling in multi-robot cells using priority rules and a genetic algorithm,” in *Proc. IEEE IECON*, 2016, pp. 692–697.
13. A. Maoudj, B. Bouzouia, A. Hentout, R. Toumi, “Task allocation and scheduling in cooperative heterogeneous robot teams: simulation results,” in *Proc. IEEE INDIN*, 2015, pp. 179–184.

Book Chapters

1. A. Maoudj, B. Bouzouia, A. Hentout, R. Toumi, “Multi-agent architecture for telerobotics of

heterogeneous robots via negotiated task allocation,” in *1^{re} Journée d’Étude en Automatique et ses Applications (1JEAA14)*, 2014, pp. 38–43.

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